

PLANT CITY, FL, USA

WMO: 720373

Lat: 28.000N

Lon: 82.164W

Elev: 154

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 92824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 36.5	(c) 40.8	(d) 20.6	(e) 15.5	(f) 51.6	(g) 27.0	(h) 21.0	(i) 54.4	(j) 18.9	(k) 70.9	(l) 17.7	(m) 71.6	(n) 3.1	(o) 10	(p) 0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 15.9	(c) 95.1	(d) 77.2	(e) 93.4	(f) 77.0	(g) 92.7	(h) 76.9	(i) 81.4	(j) 87.7	(k) 80.9	(l) 87.4	(m) 80.2	(n) 86.9	(o) 5.8	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	153.9	82.8	79.2	152.6	82.8	78.8	150.4	82.6	44.9	88.0	44.4	87.2	43.8	88.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.9	(b) 14.7	(c) 12.4	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
			(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.2	61.2	65.4	67.7	74.0	79.2	82.3	83.6	84.1	82.3	76.4	68.4	66.0
	DBStd	9.78	8.85	8.70	7.03	4.74	3.62	2.44	2.12	2.00	2.17	5.74	7.23	8.35
	HDD50	27	17	5	0	0	0	0	0	0	0	0	0	5
	HDD65	487	177	95	59	4	0	0	0	0	0	7	49	96
	CDD50	8879	364	438	548	720	906	968	1040	1056	970	817	552	500
	CDD65	3862	59	107	141	274	441	518	575	591	520	360	151	125
	CDH74	37137	394	790	1199	2566	4536	5287	6015	6281	5138	3112	1053	766
	CDH80	15151	53	176	297	945	2099	2438	2681	2749	2188	1136	241	148
(14) Wind	WSAvg	5.0	5.5	5.4	6.0	5.9	5.6	4.6	3.7	3.9	4.2	4.9	4.9	5.2
(15) Precipitation	PrecAvg	53.9	2.7	2.6	3.2	2.6	2.9	8.7	8.2	8.6	7.6	2.6	1.8	2.2
	PrecMax	68.8	6.5	8.7	8.5	7.7	8.8	15.9	12.0	14.9	16.5	6.2	6.8	12.1
	PrecMin	38.1	0.4	0.3	0.4	0.3	0.4	2.3	4.7	4.5	2.6	0.2	0.2	0.1
	PrecStd	8.0	1.6	1.9	2.5	1.5	2.0	3.3	2.0	2.6	3.4	1.5	1.4	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.0	87.5	87.9	92.7	95.5	96.8	95.5	95.5	94.7	91.5	88.3	86.2
		MCWB	69.8	70.4	70.6	72.8	72.5	76.4	77.9	78.8	77.4	75.4	75.6	73.4
	2%	DB	81.4	84.2	85.8	90.1	93.1	94.7	94.8	94.8	92.8	90.3	85.6	83.7
		MCWB	68.7	69.6	68.7	70.8	72.2	76.6	77.8	78.8	77.2	75.5	73.4	72.2
	5%	DB	79.0	81.9	83.6	88.0	91.1	92.6	93.0	93.0	91.1	88.3	82.3	81.5
		MCWB	67.0	68.3	68.3	70.3	72.5	76.7	77.7	78.8	76.7	75.0	70.5	70.8
	10%	DB	75.1	79.4	81.3	85.8	90.1	90.8	91.0	91.1	90.2	86.2	80.8	79.0
		MCWB	65.5	66.5	66.2	69.2	72.3	76.6	78.0	78.8	76.7	73.6	69.0	68.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.7	74.2	75.2	77.5	78.6	81.5	82.0	82.4	81.6	80.0	78.0	76.4
		MCDB	79.6	79.2	81.9	84.8	85.3	87.8	88.1	87.8	87.3	84.8	84.6	81.7
	2%	WB	71.6	72.5	73.1	75.4	77.4	80.5	81.2	81.4	80.8	78.6	75.7	74.8
		MCDB	77.2	79.1	78.8	83.4	85.5	87.2	87.6	88.1	86.7	84.6	81.4	80.3
	5%	WB	69.8	71.3	71.3	73.8	76.4	79.5	80.4	80.9	79.8	77.8	73.8	73.1
		MCDB	74.9	78.5	77.9	81.8	84.6	86.5	87.3	87.7	85.7	84.2	79.2	78.1
	10%	WB	68.1	69.8	69.5	72.5	75.4	78.8	79.7	80.2	79.0	76.7	72.0	71.8
		MCDB	73.5	76.5	76.3	80.6	83.8	85.9	86.9	87.2	85.0	83.0	77.6	77.1
(35) Mean Daily Temperature Range	5% DB	MDBR	21.6	21.9	22.9	22.1	21.3	17.9	16.5	15.9	16.4	18.8	20.8	20.5
		MCDBR	22.2	21.6	23.2	23.7	22.7	20.0	18.1	17.3	17.3	17.8	20.0	19.9
		MCWBR	11.7	9.3	8.4	7.8	6.8	6.2	5.7	5.6	5.7	6.4	9.4	9.9
	5% WB	MCDBR	19.2	18.2	19.5	20.2	19.0	17.1	16.1	15.8	15.4	15.3	16.6	17.8
		MCWBR	12.2	9.7	9.6	8.9	6.8	6.5	6.1	5.8	5.8	6.2	8.4	10.3
(40) Clear-Sky Solar Irradiance	taub		0.338	0.351	0.365	0.386	0.422	0.448	0.472	0.453	0.419	0.388	0.352	0.338
	taud		2.519	2.494	2.443	2.353	2.307	2.312	2.246	2.316	2.388	2.449	2.496	2.546
	Ebn at Noon		288	293	294	289	277	268	261	266	273	276	280	283
	Edh at Noon		29	32	36	40	42	42	44	41	37	33	29	27
(44) All-Sky Solar Radiation	RadAvg		1118	1342	1691	2014	2117	1897	1824	1757	1601	1477	1209	1030
	RadStd		80	83	133	99	127	157	115	112	96	94	62	82

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-7	-81	+284	+202

Nomenclature: See separate page